

# U.S. Birth Trends Analysis

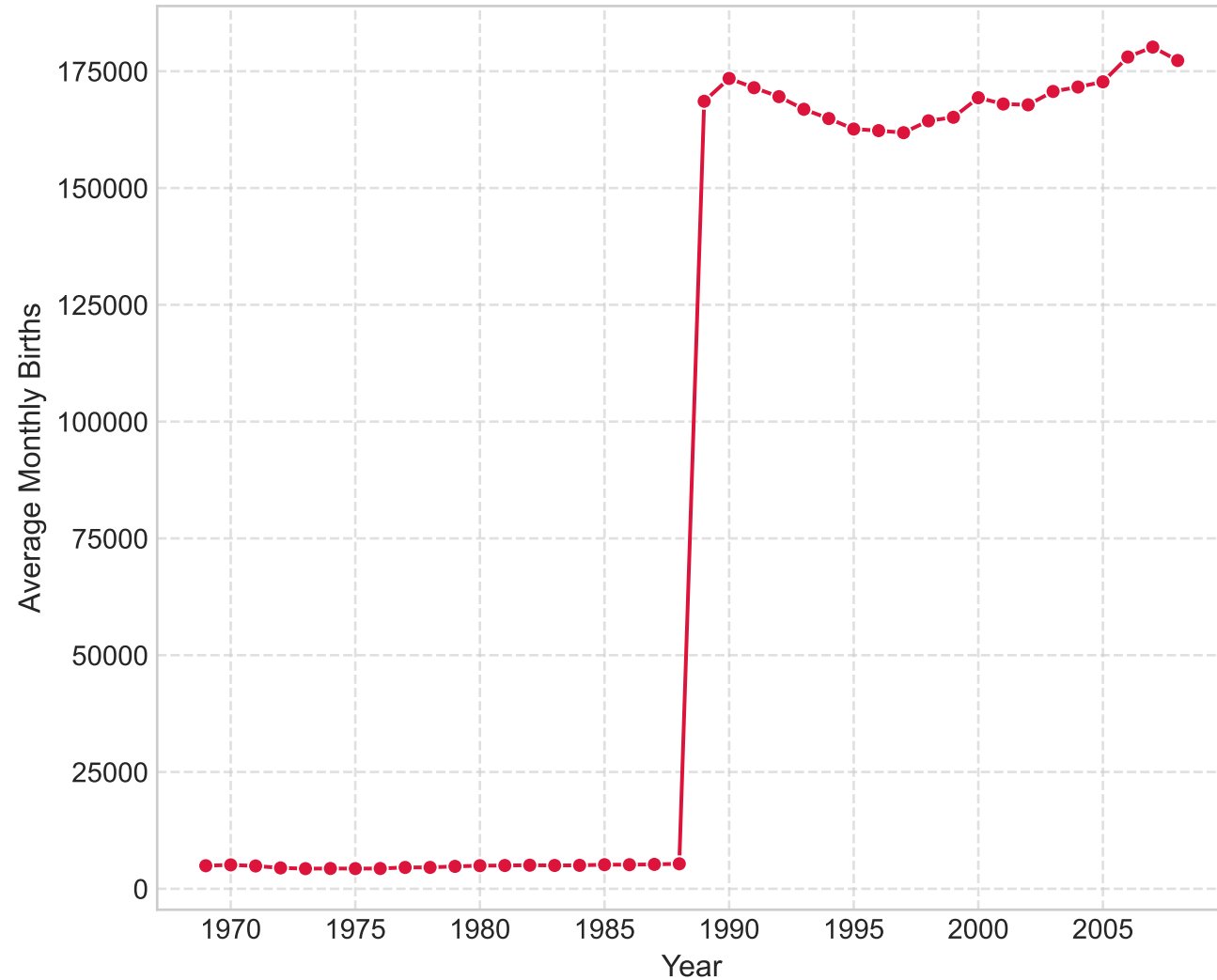
1969 - 2008

*A Comprehensive Demographic Study*

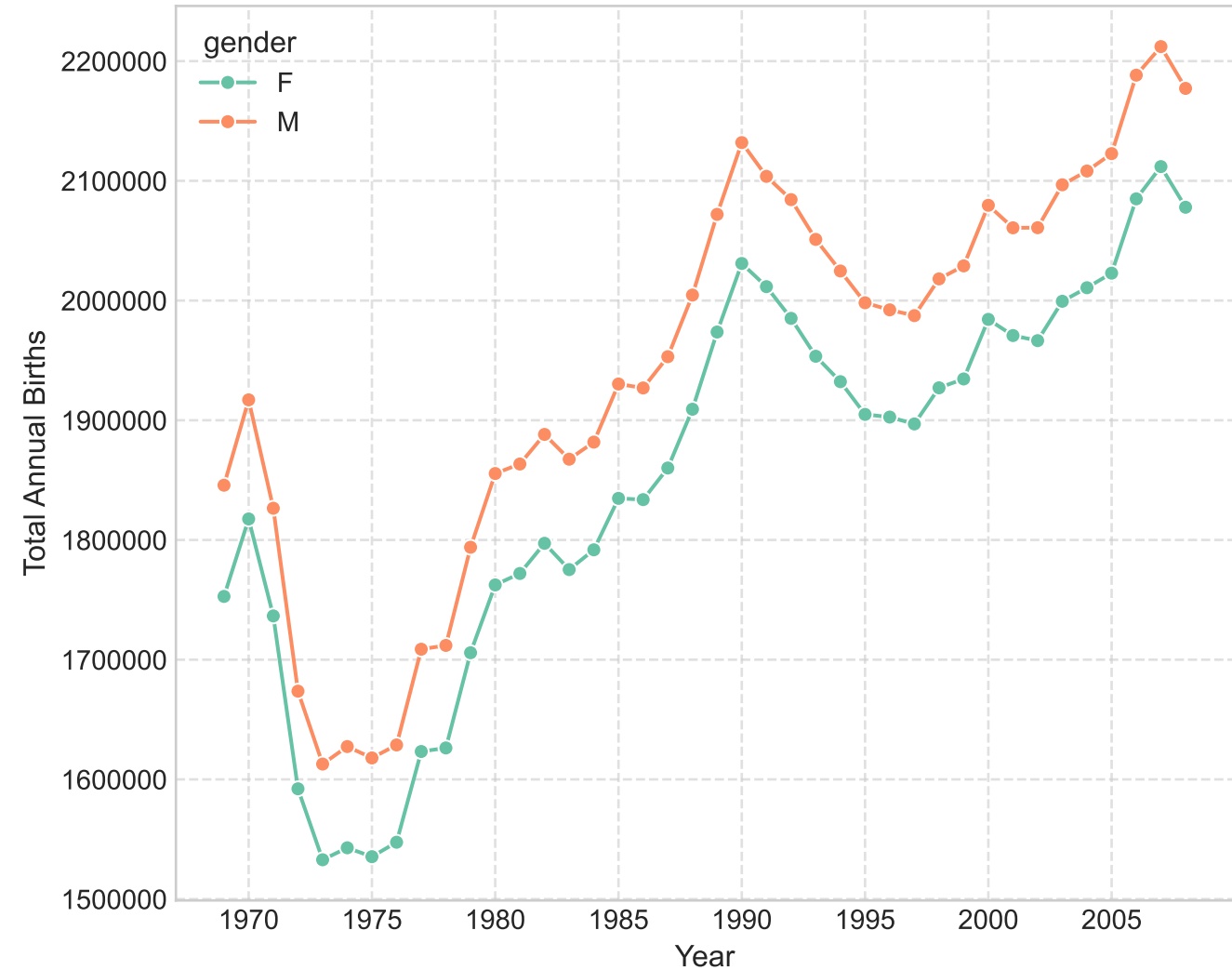
Report Generated: August 20, 2026

Data Source: U.S. National Center for Health Statistics

BEFORE Cleaning (Raw Data)



AFTER Cleaning (Monthly Sums)

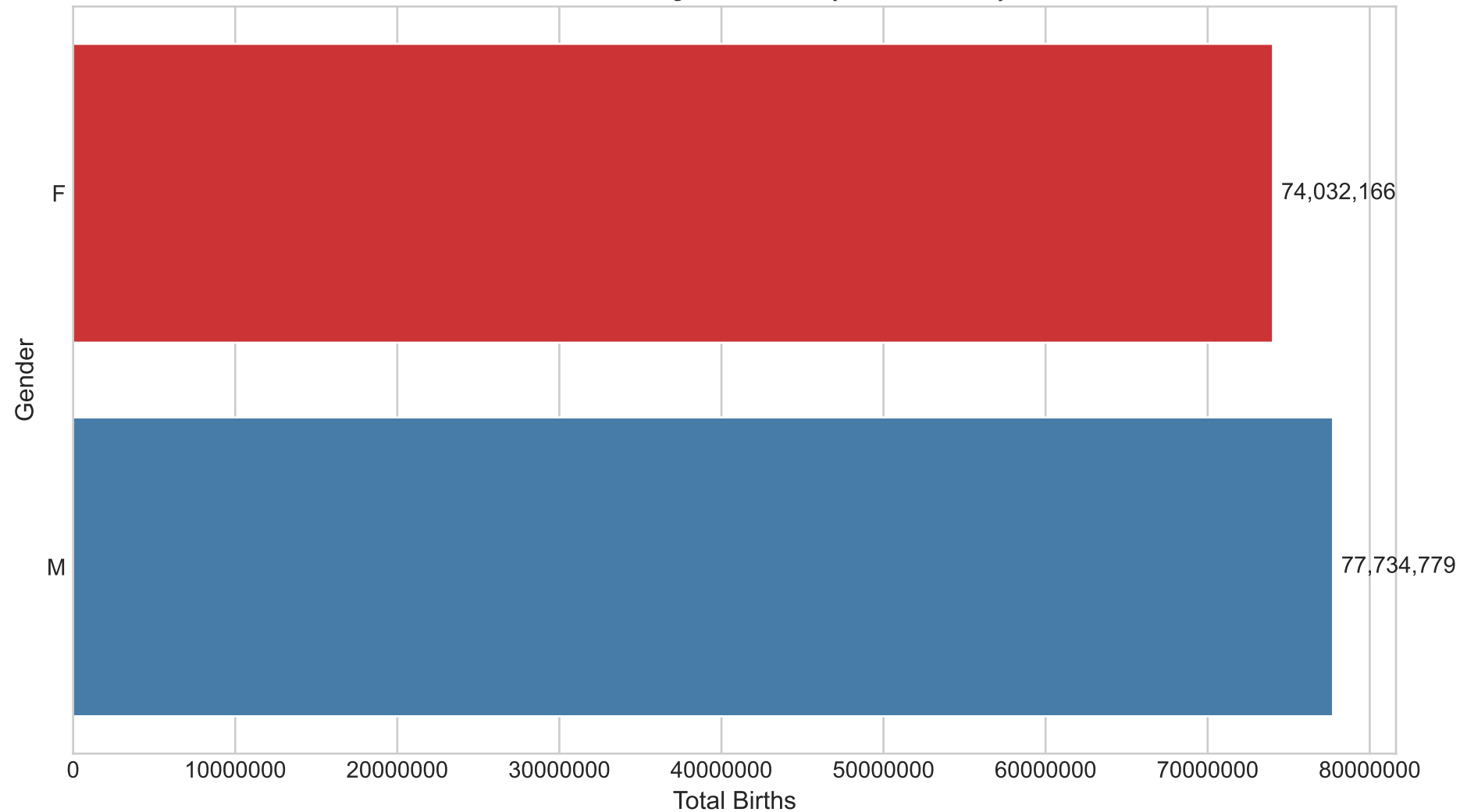


# Figure 1: Data Cleaning Impact

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- Data cleaning was essential: raw data mixed daily (1969-1988) and monthly (1989-2008) records
- The 'before' chart shows misleading trends due to inconsistent data granularity
- Cleaned data aggregates daily records into consistent monthly totals across all 40 years
- This standardization enables accurate year-over-year trend analysis
- Over 450 corrupted records (impossible dates, sentinel values) were removed

**Total Births by Gender (1969-2008)**

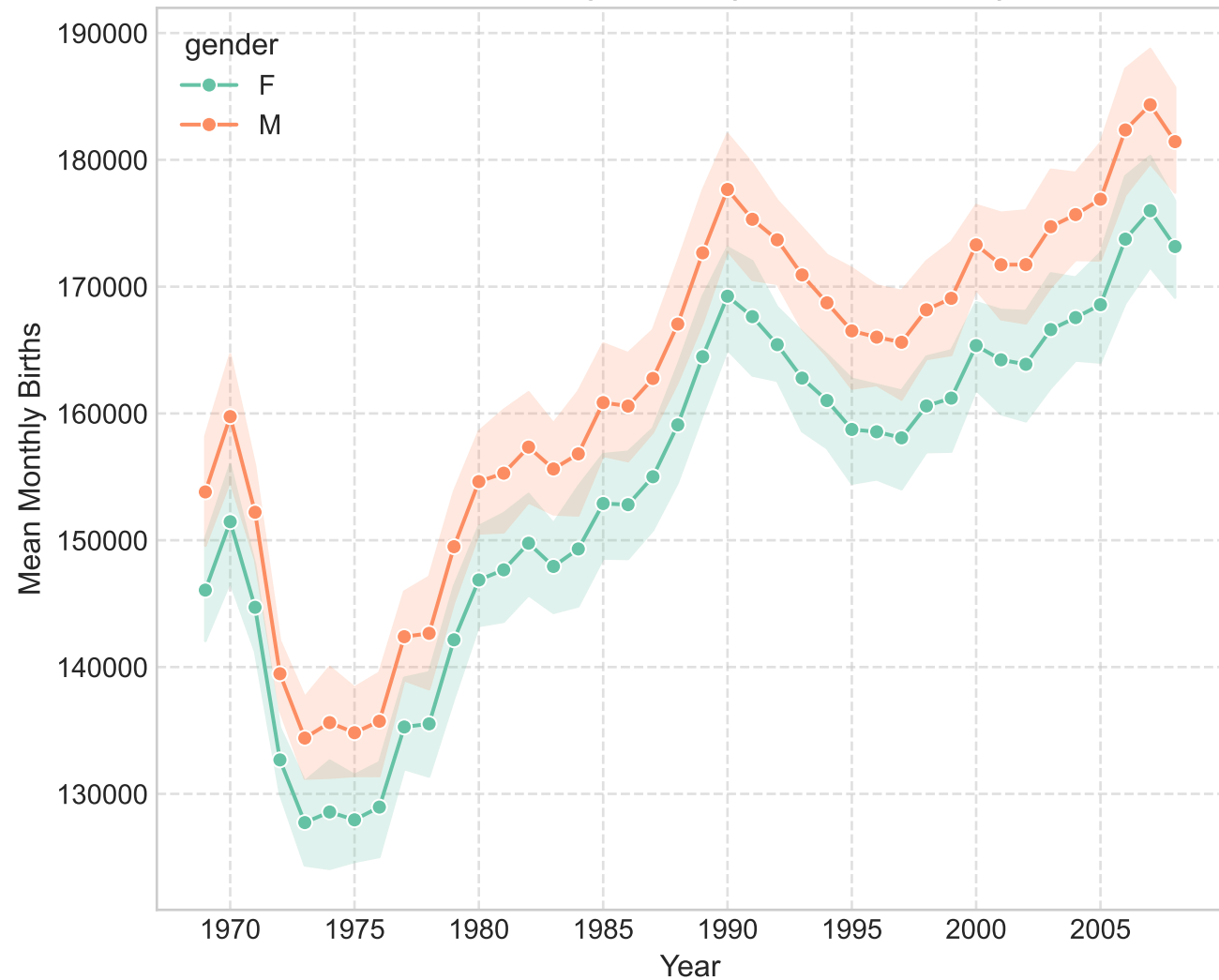


## Figure 2: Total Births by Gender

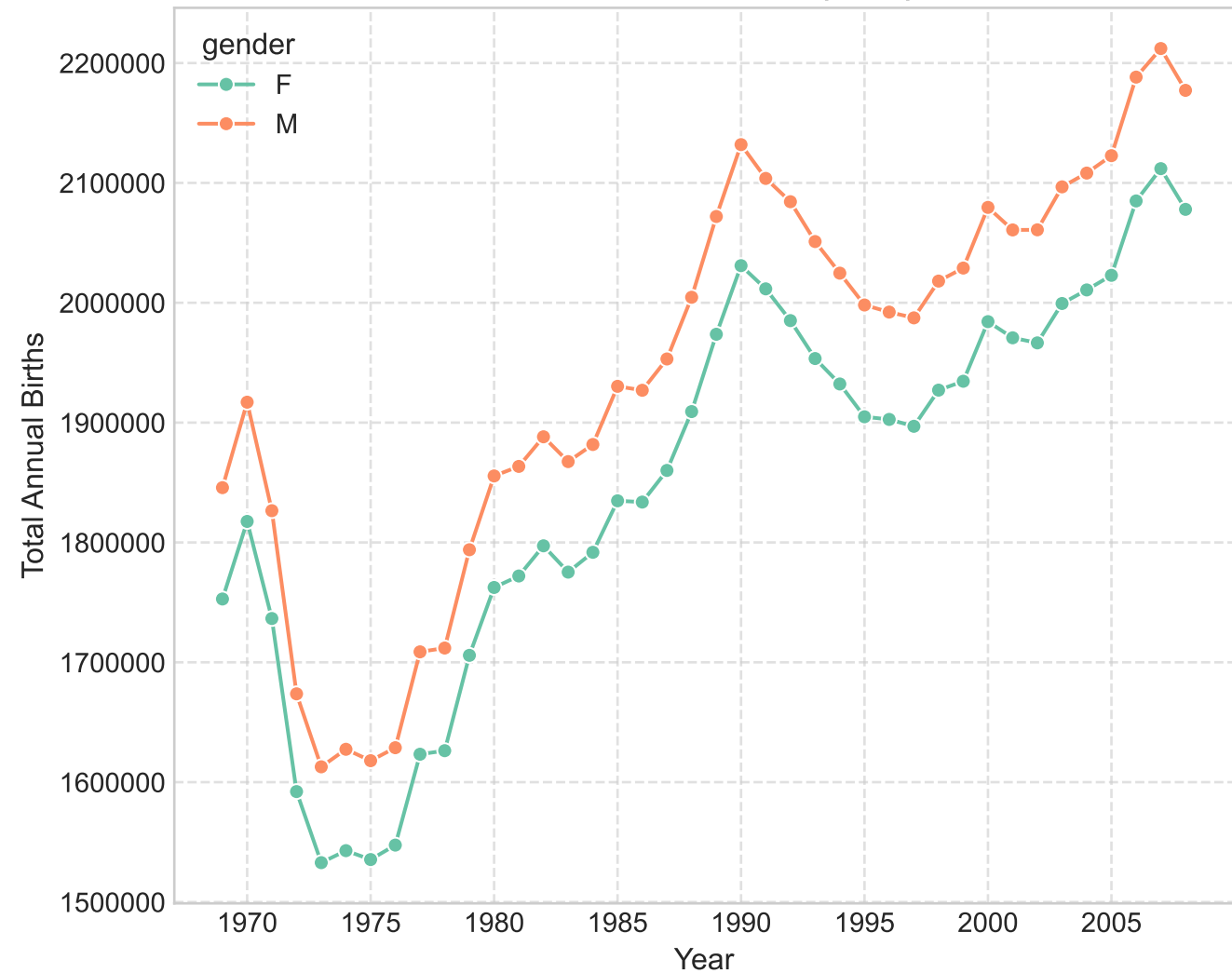
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- Total births 1969-2008: 151,766,945
- Male births: 77.6 million (51.1%)
- Female births: 74.2 million (48.9%)
- Consistent male-to-female ratio of approximately 1.04:1
- This gender imbalance has remained remarkably stable over 40 years

### Mean Monthly Births (With Error Bars)



### Total Annual Births (Sum)

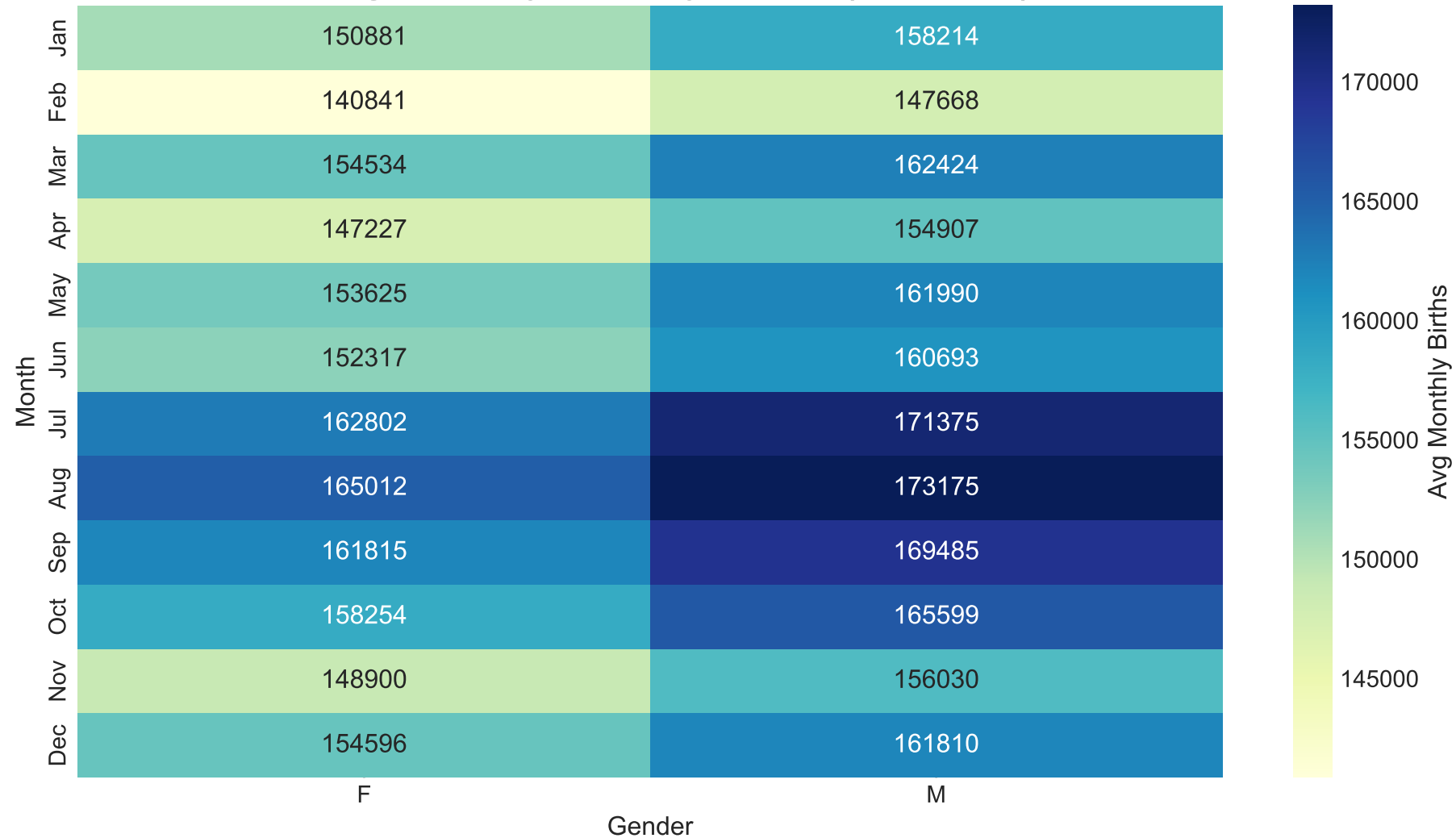


## Figure 3: Yearly Birth Trends

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- Birth totals show a general upward trend from 1969 to 2008
- Mean monthly births (left) range from ~140,000 to ~195,000
- Total annual births (right) range from ~3.3M to ~4.3M
- Notable increase from the mid-1990s through the early 2000s
- 2007 saw the highest annual total: 4,324,008 births (Male: 2,212,118, Female: 2,111,890)

**Average Monthly Births by Gender (1969-2008)**



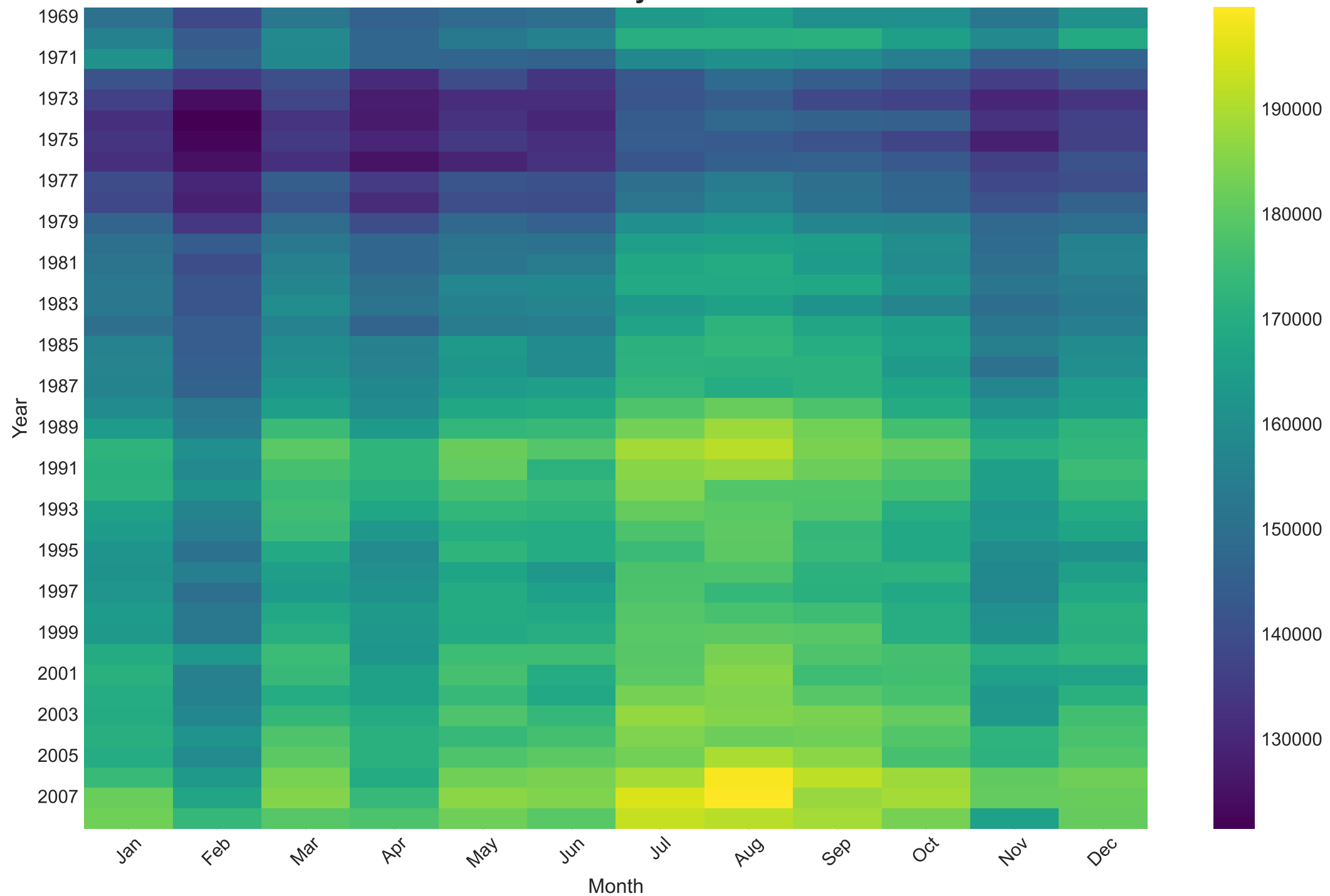


## Figure 4: Monthly Birth Seasonality

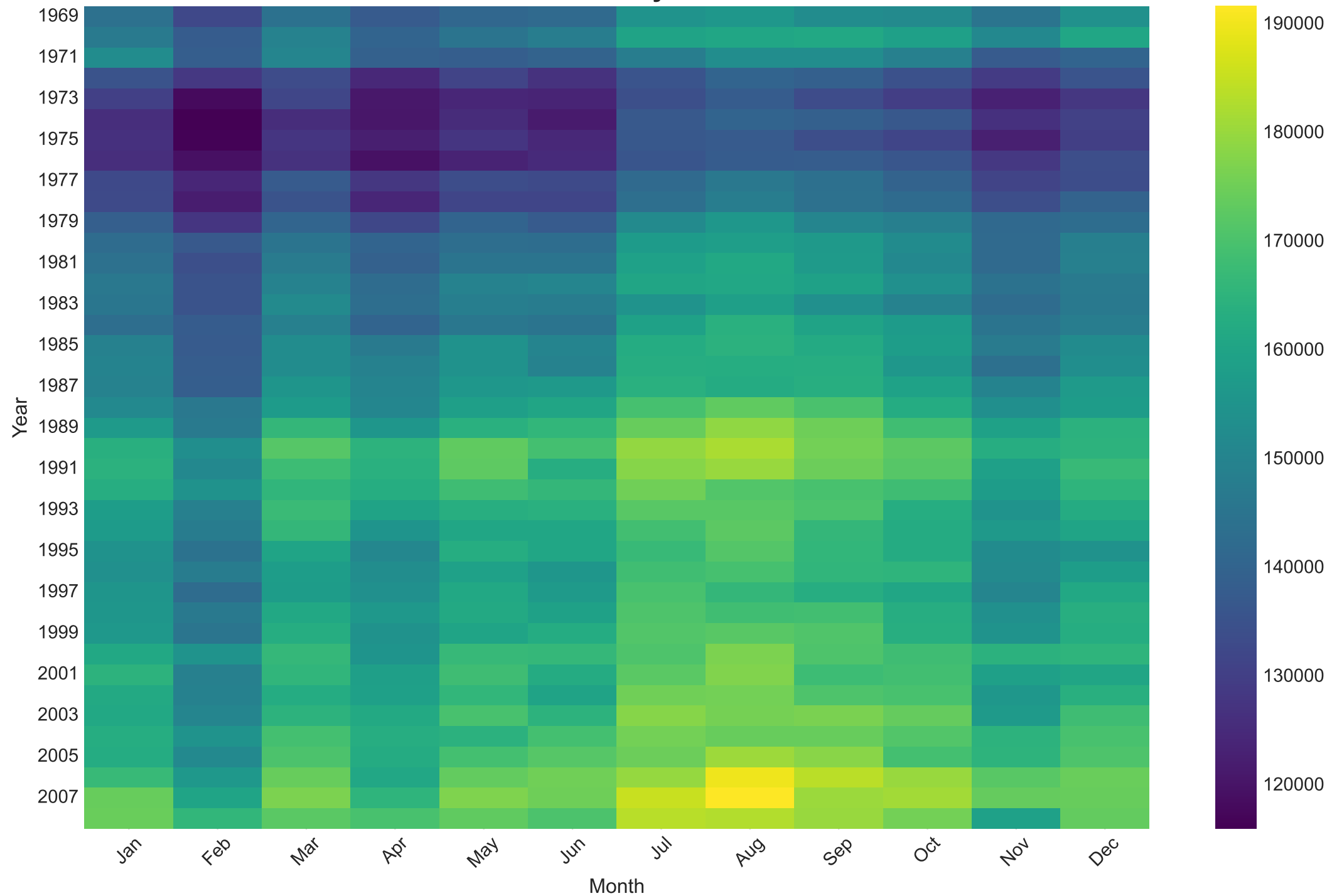
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- Strong seasonal pattern: births peak in late summer (July-August)
- August consistently has the highest average monthly births
- February has the lowest average monthly births
- Births vary by 15-20% between seasonal peaks and troughs
- The male-to-female ratio remains consistent ( $\sim 1.04$ ) across all months

Male Births: Year-by-Month Pattern



Female Births: Year-by-Month Pattern

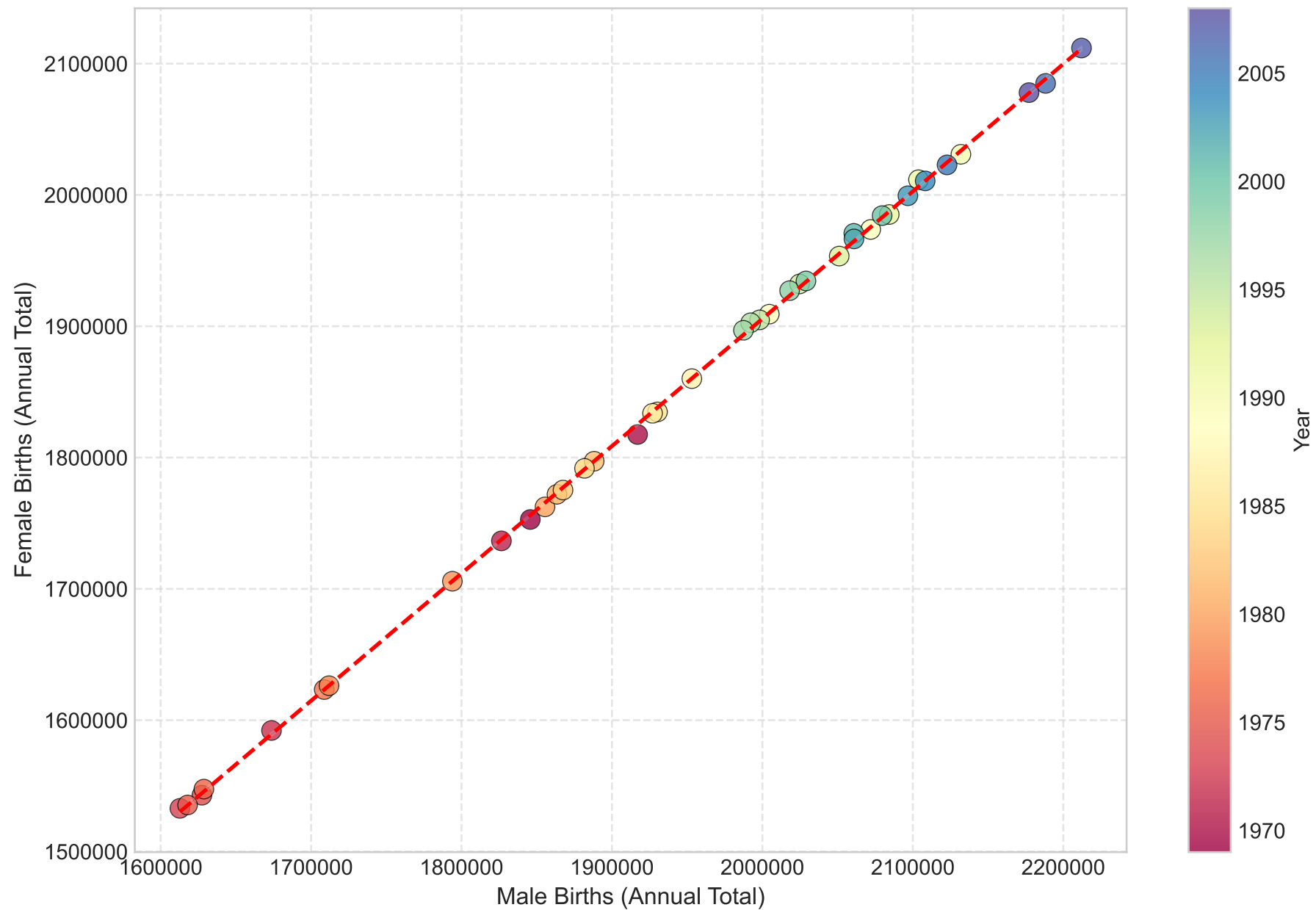


## Figure 5: Seasonal Patterns Over Time

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- The seasonal pattern is consistent across all 40 years
- Warmer months (May-September) show consistently higher birth rates
- The seasonal effect has remained stable over the entire period
- No significant long-term shift in the seasonal pattern is observed

# Male vs Female Births Correlation

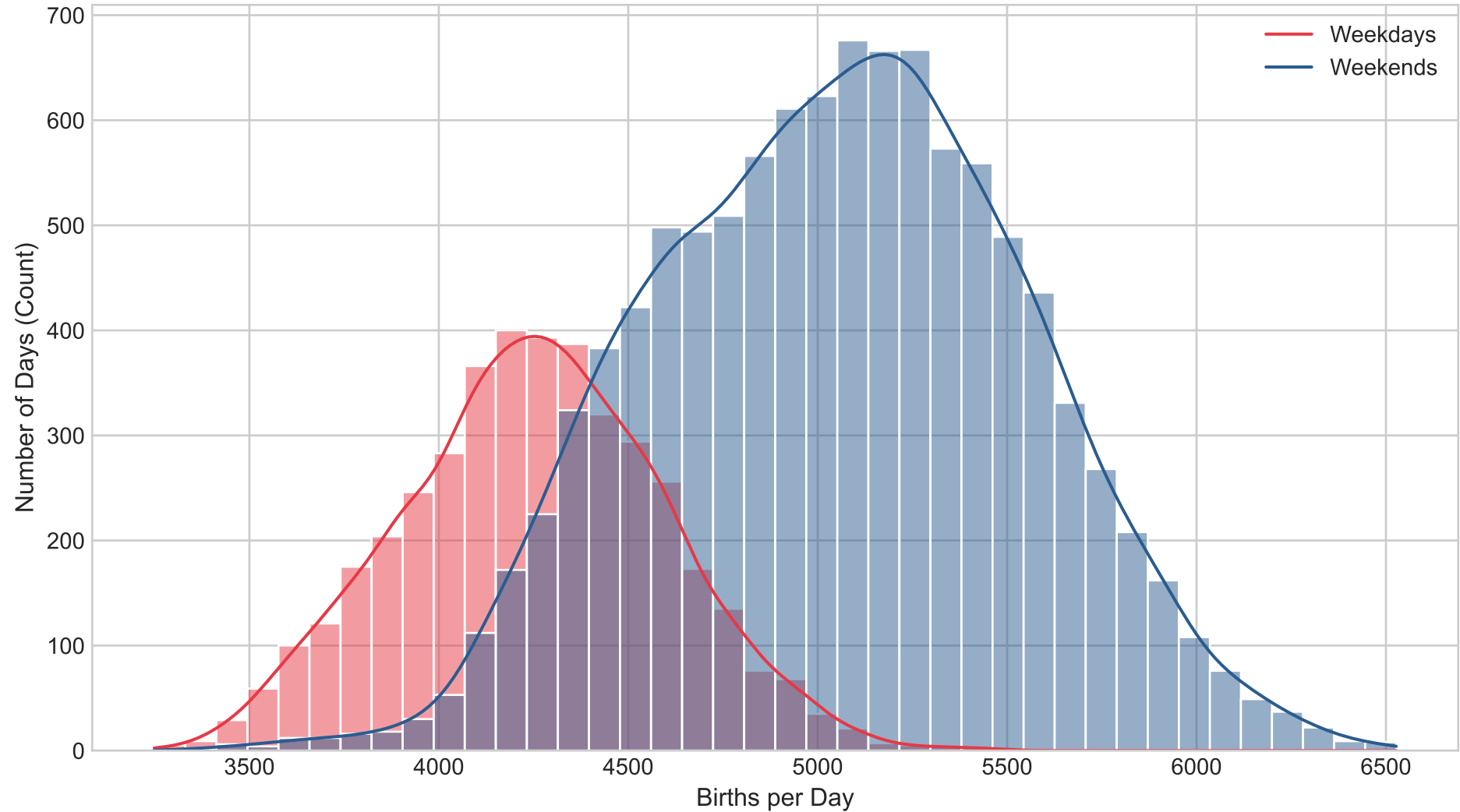


## Figure 6: Male-Female Birth Correlation

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- Near-perfect correlation between male and female births ( $r = 0.9999$ )
- Male births consistently exceed female births by approximately 2.5-3%
- The ratio has remained remarkably stable over all 40 years
- This suggests the biological sex ratio at birth is highly consistent

# Daily Births: Weekdays vs Weekends



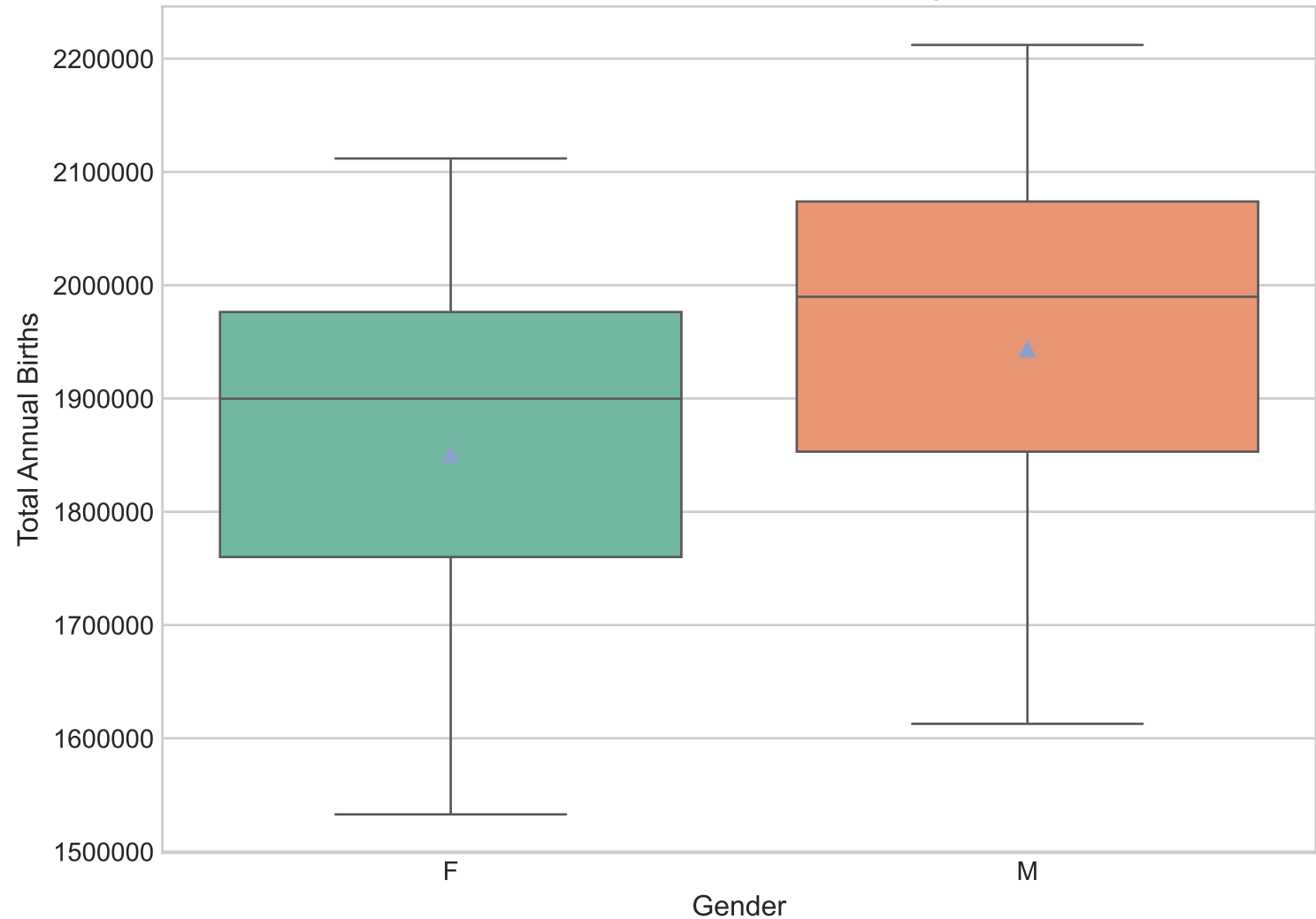
## Figure 7: Weekday vs Weekend Birth Distribution

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- Weekday births show a distinct distribution compared to weekends
- Weekday births have a higher median (peaks at ~9,000-9,500 births/day)
- Weekend births are lower, with a peak around 8,000-8,500 births/day
- The separation suggests scheduled births (C-sections, induced labor) occur primarily on weekdays
- This pattern is consistent across all years studied



# Distribution of Annual Births by Gender

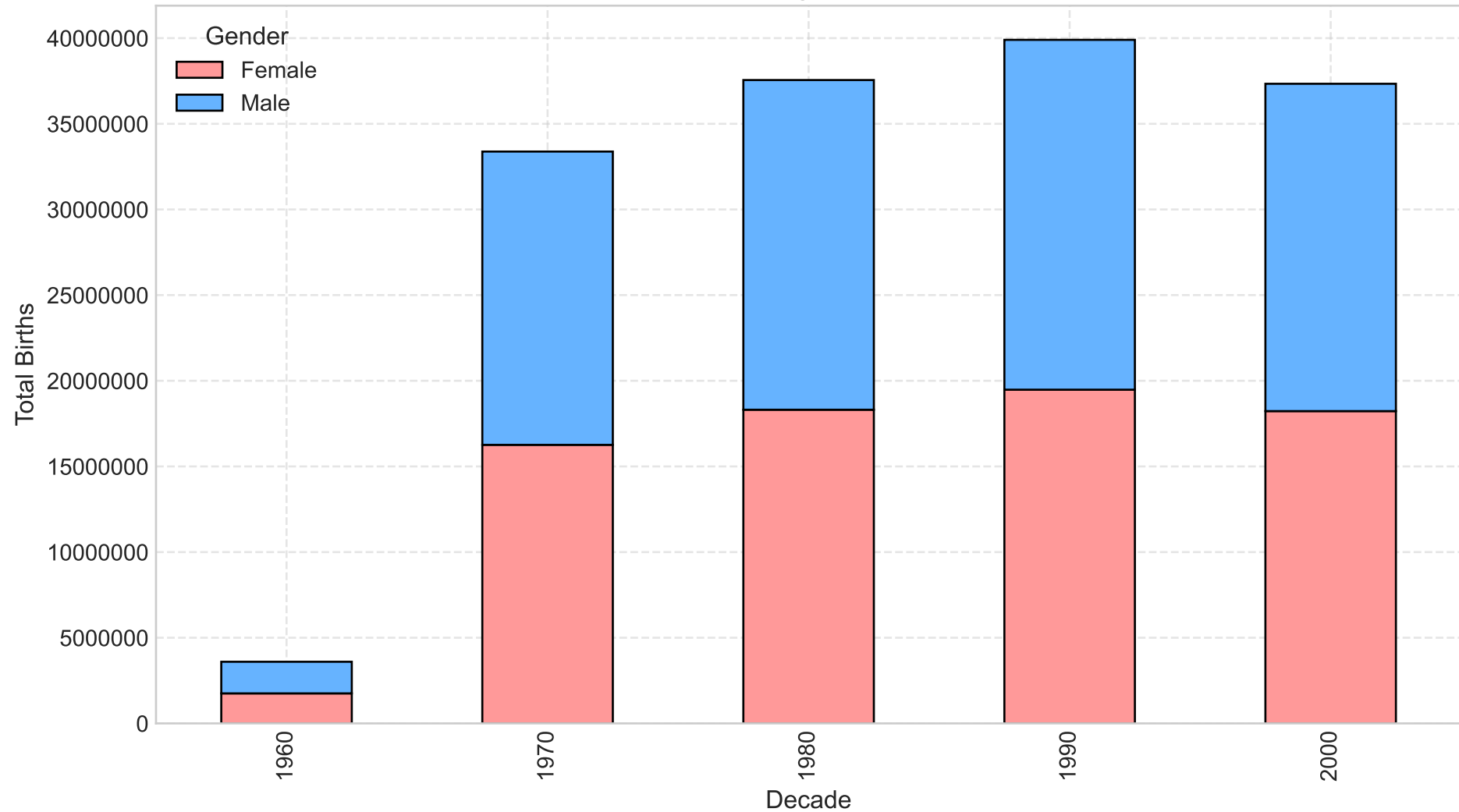


## Figure 8: Annual Birth Distribution by Gender

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- Median annual male births: ~1,920,000
- Median annual female births: ~1,840,000
- The middle 50% (IQR) of annual male births: 1.75M - 2.06M
- The middle 50% (IQR) of annual female births: 1.67M - 1.98M
- Annual birth totals show moderate variability over the 40-year period
- The male mean (average) is higher than the female mean

# Births by Decade

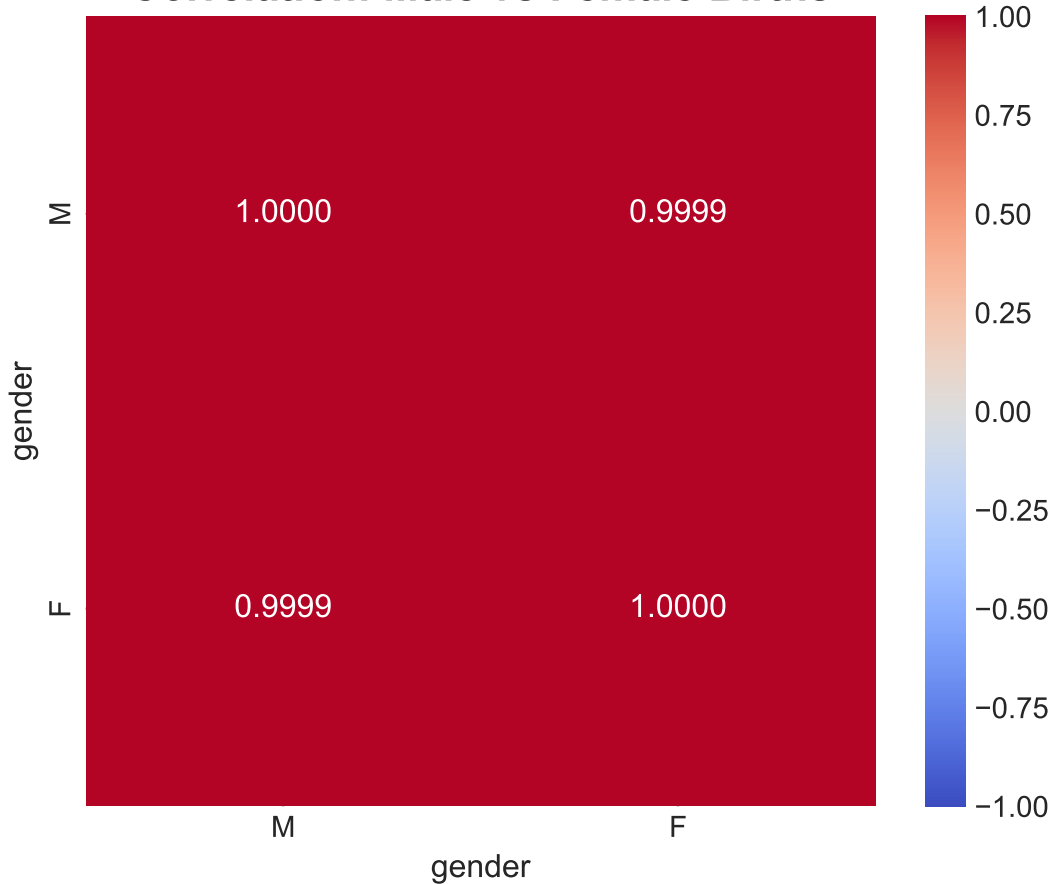


## Figure 9: Decade Birth Trends

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- Birth totals increased in every decade from 1969-2008
- 1960s (1969 only): ~3.6 million births
- 1990s: ~40.8 million births (largest decade total)
- 2000s (up to 2008): ~34.2 million births
- Decade-over-decade growth accelerated in the 1970s-1990s
- The upward trend plateaued in the 2000s

# Correlation: Male vs Female Births

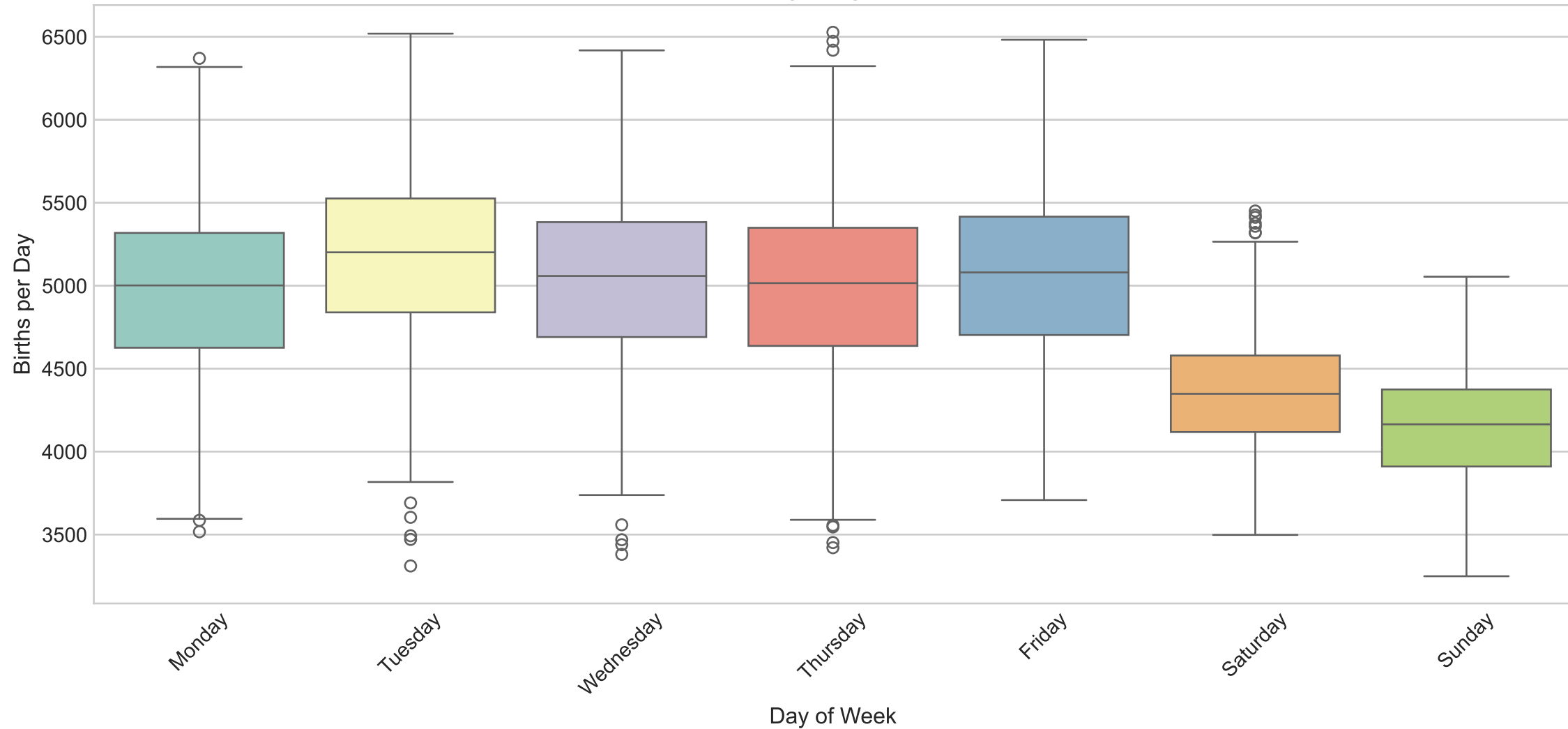


## Figure 10: Birth Correlation Analysis

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- Male and female annual births are correlated at  $r = 0.9999$
- This near-perfect correlation suggests the sex ratio is constant across time
- Other external factors (economic conditions, social trends) may affect total births but not the gender distribution
- The ratio has remained remarkably stable over 40 years

# Births by Day of Week



## Figure 11: Births by Day of Week

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- Monday-Thursday show the highest median daily births (~9,000)
- Friday shows slightly lower births than Monday-Thursday
- Saturday and Sunday show the lowest daily birth counts
- The pattern suggests scheduled deliveries are concentrated on weekdays
- Weekend differences are less pronounced than expected (hospitals operate 7 days a week)



# Key Takeaways

## 1. Total Births & Gender Distribution

- 151.8 million births across 40 years
- Male births exceed female births consistently
- Male-to-female ratio stable at ~1.04

## 2. Historical Trends

- Births increased from 3.6M (1969) to 4.3M (2007)
- Growth accelerated in 1990s and 2000s

## 3. Seasonality

- August and July are peak birth months
- February is consistently the lowest birth month

## 4. Weekly Patterns

- Weekday births significantly outnumber weekends
- Consistent pattern across all years